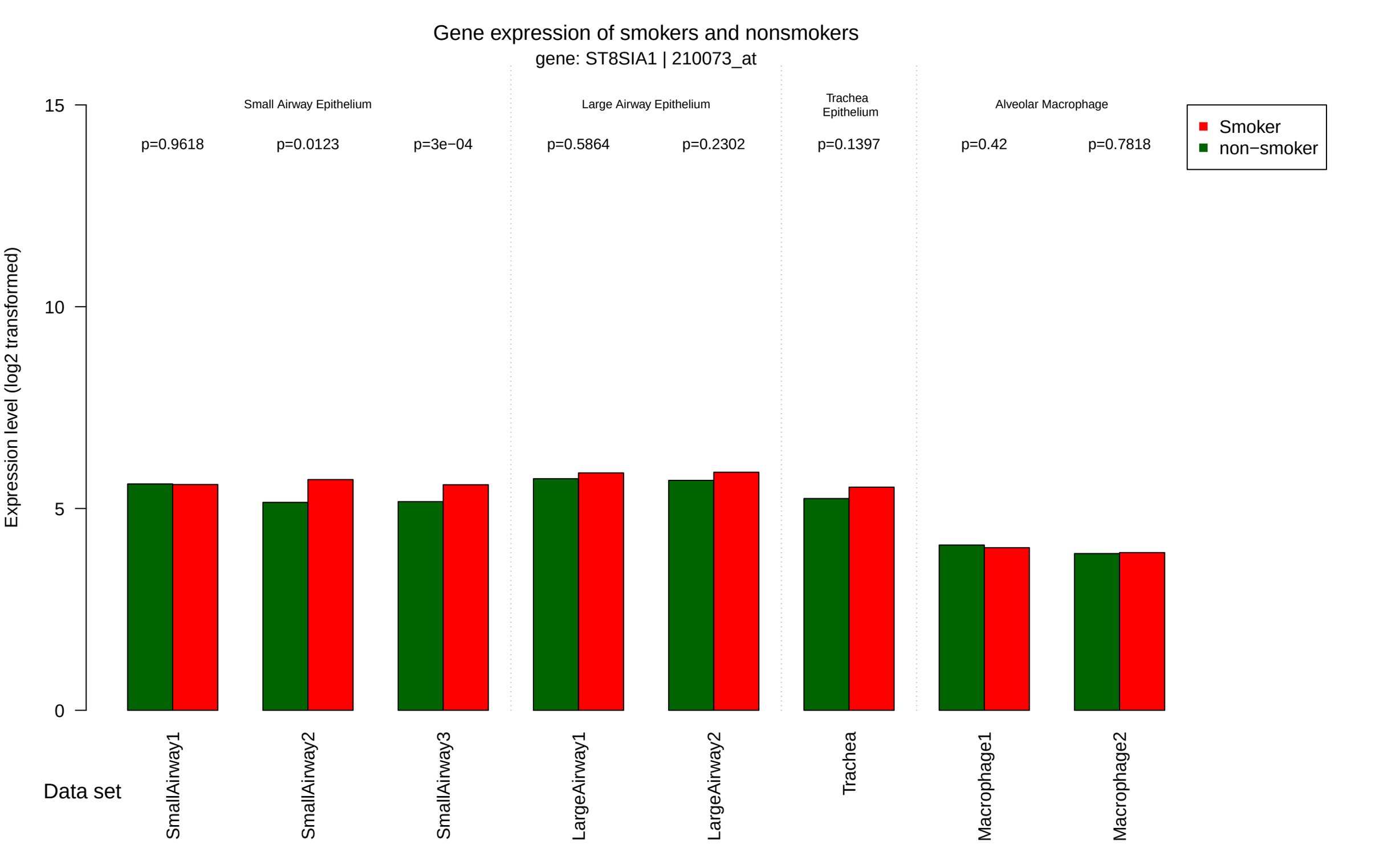
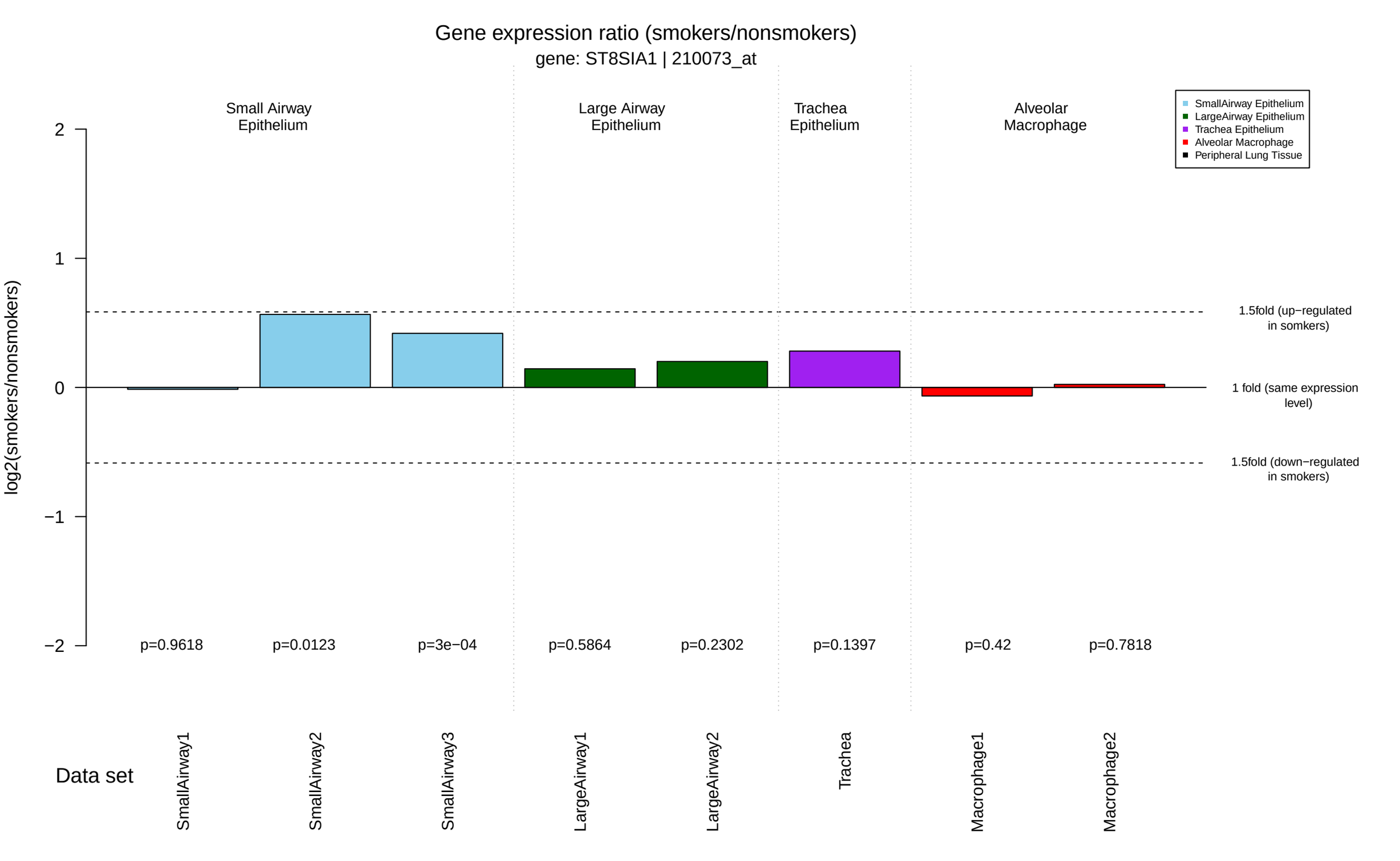
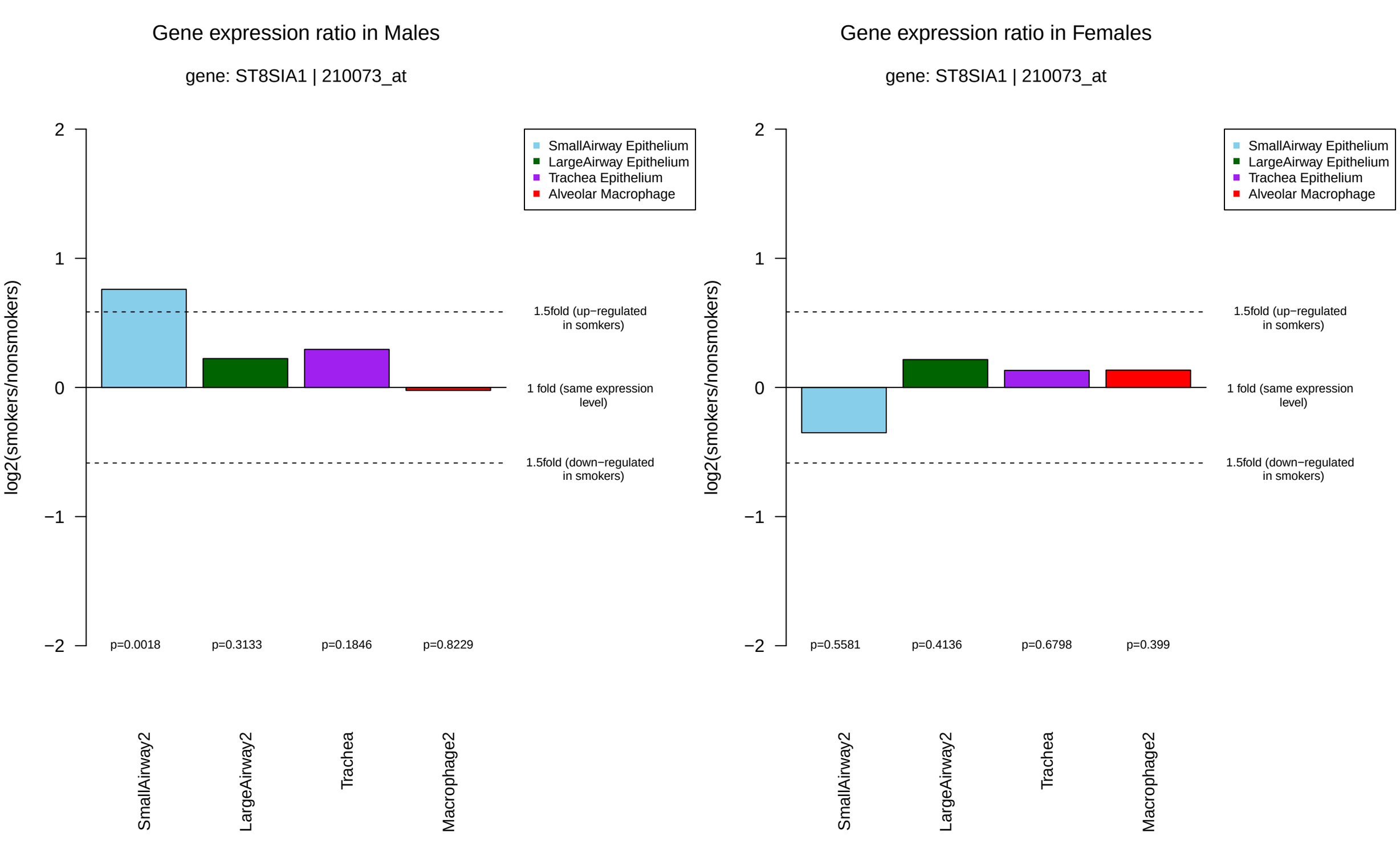
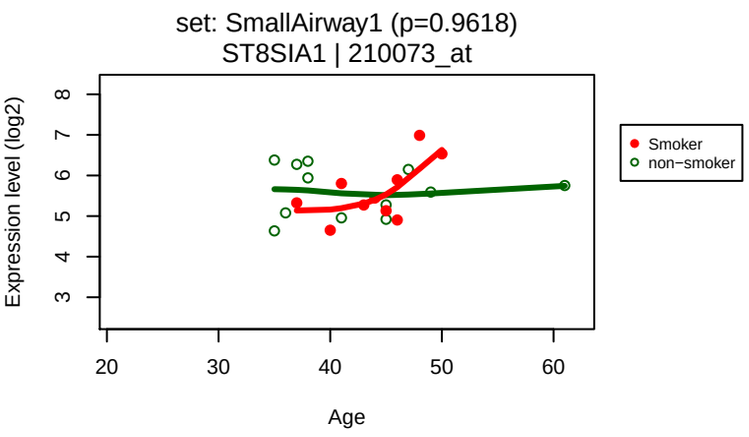
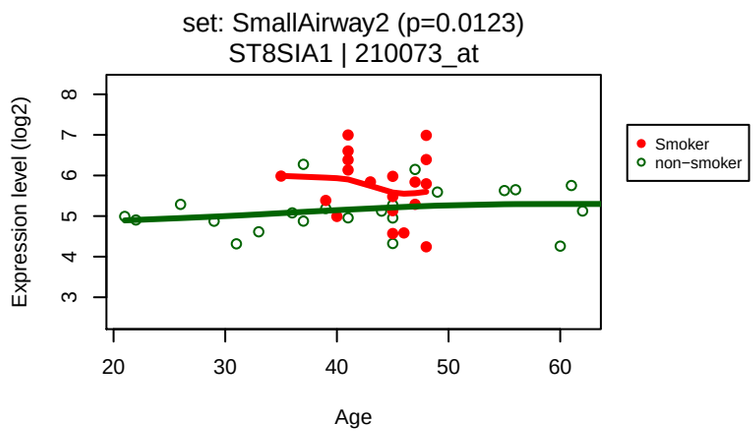
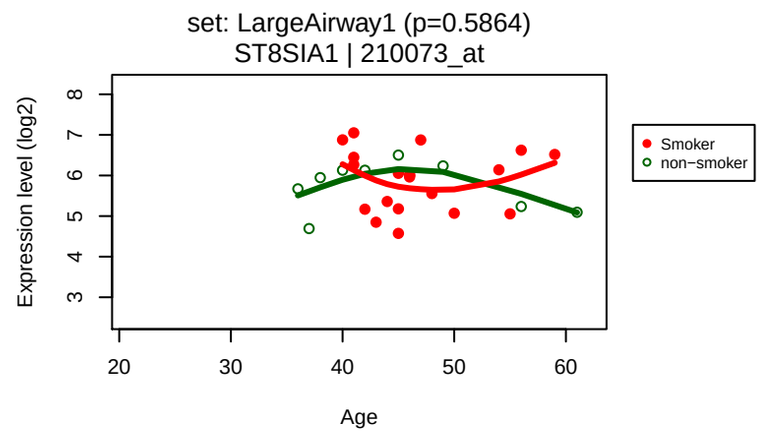

Gene expression of smokers and nonsmokers
gene: ST8SIA1 | 210073_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9618
p=0.0123
p=3e−04
p=0.5864
p=0.2302
p=0.1397
p=0.42
p=0.7818
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

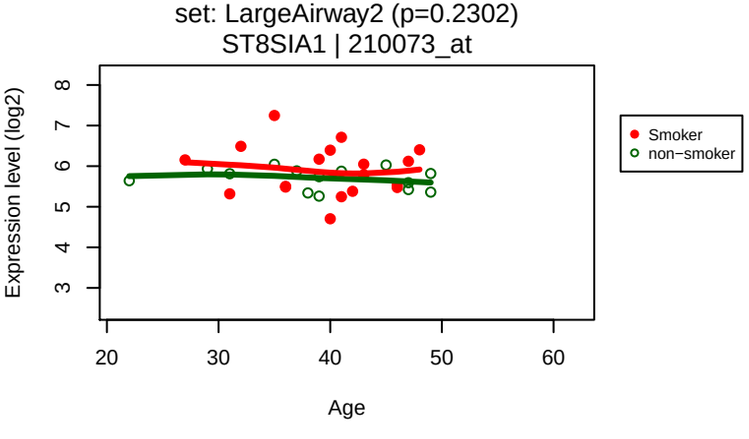

Gene expression ratio (smokers/nonsmokers)
gene: ST8SIA1 | 210073_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9618
p=0.0123
p=3e−04
p=0.5864
p=0.2302
p=0.1397
p=0.42
p=0.7818
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

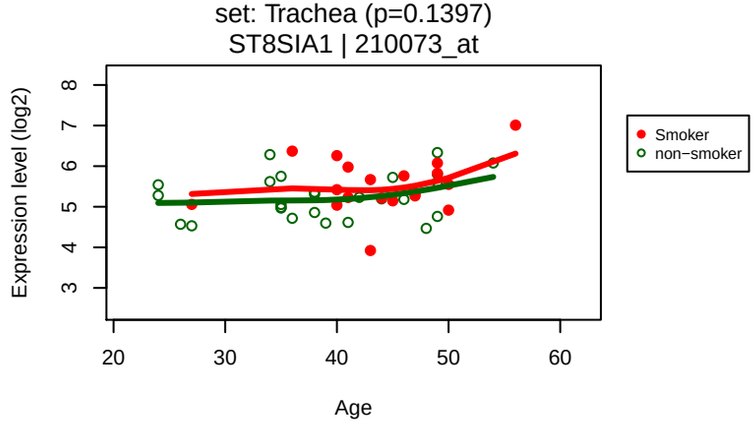

Gene expression ratio in Males
gene: ST8SIA1 | 210073_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0018
p=0.3133
p=0.1846
p=0.8229
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ST8SIA1 | 210073_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5581
p=0.4136
p=0.6798
p=0.399
SmallAirway2
LargeAirway2
Trachea
Macrophage2

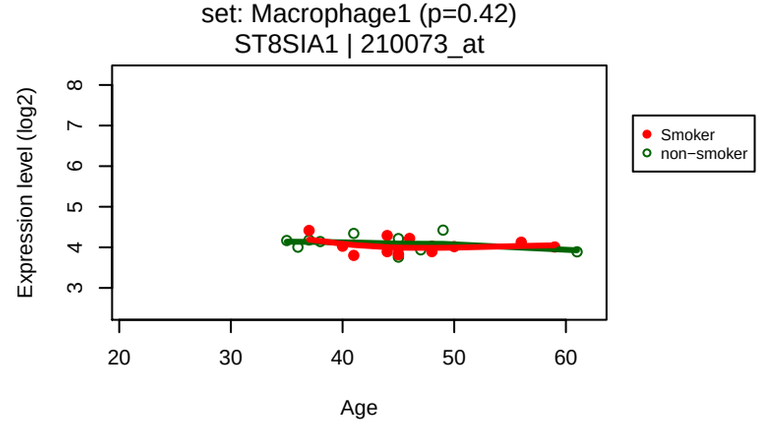

set: SmallAirway1 (p=0.9618)
ST8SIA1 | 210073_at
Expression level (log2)
Age
Smoker
non−smoker

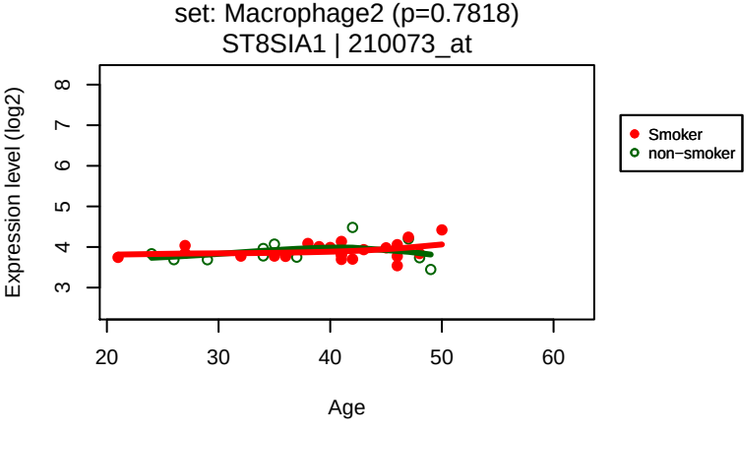

set: SmallAirway2 (p=0.0123)
ST8SIA1 | 210073_at
Expression level (log2)
Age
Smoker
non−smoker

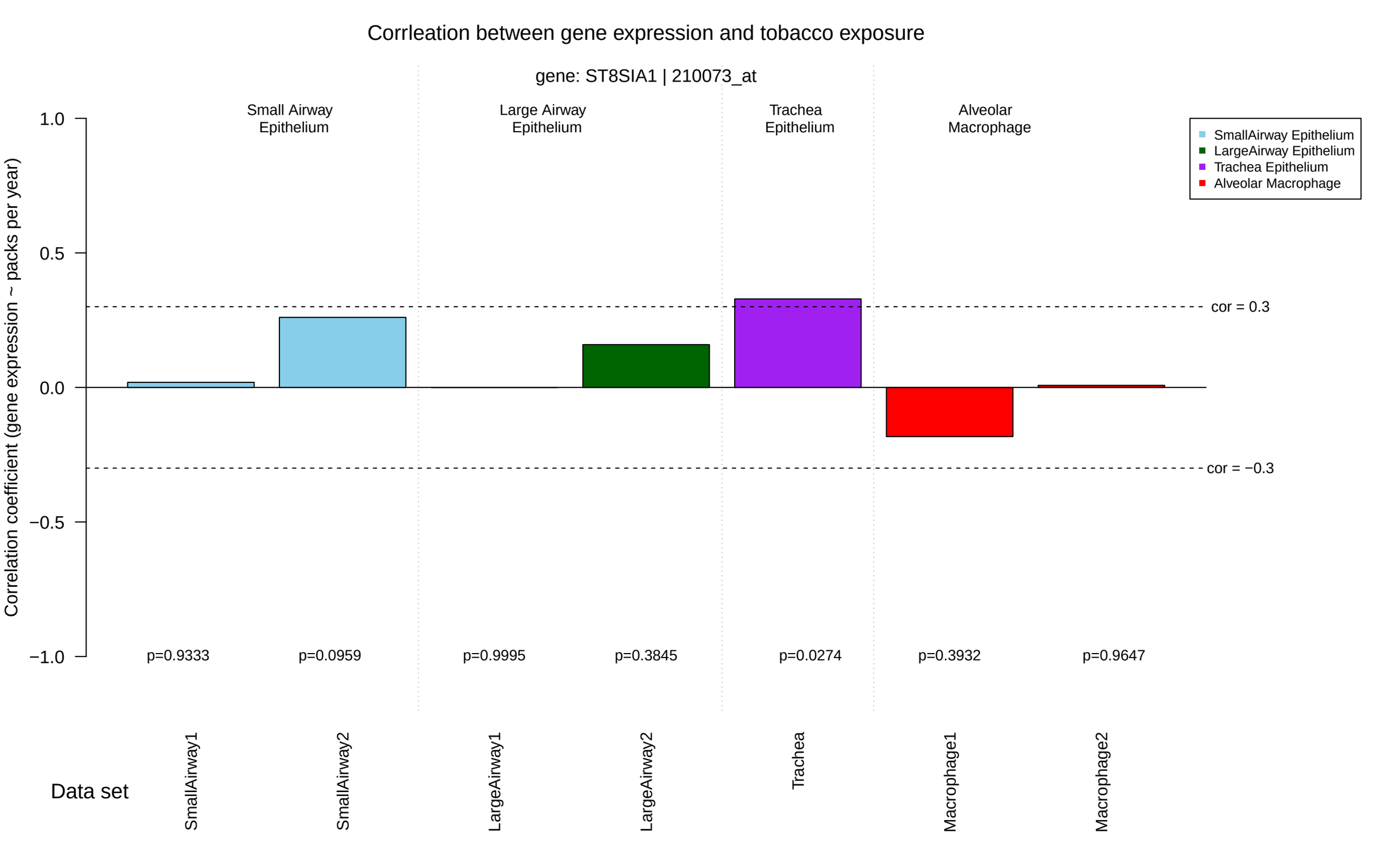

set: LargeAirway1 (p=0.5864)
ST8SIA1 | 210073_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2302)
ST8SIA1 | 210073_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1397)
ST8SIA1 | 210073_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.42)
ST8SIA1 | 210073_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7818)
ST8SIA1 | 210073_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ST8SIA1 | 210073_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9333
p=0.0959
p=0.9995
p=0.3845
p=0.0274
p=0.3932
p=0.9647
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2